UCBM
UNIVERSITA' CAMPUS BIO-MEDICO DI ROMA

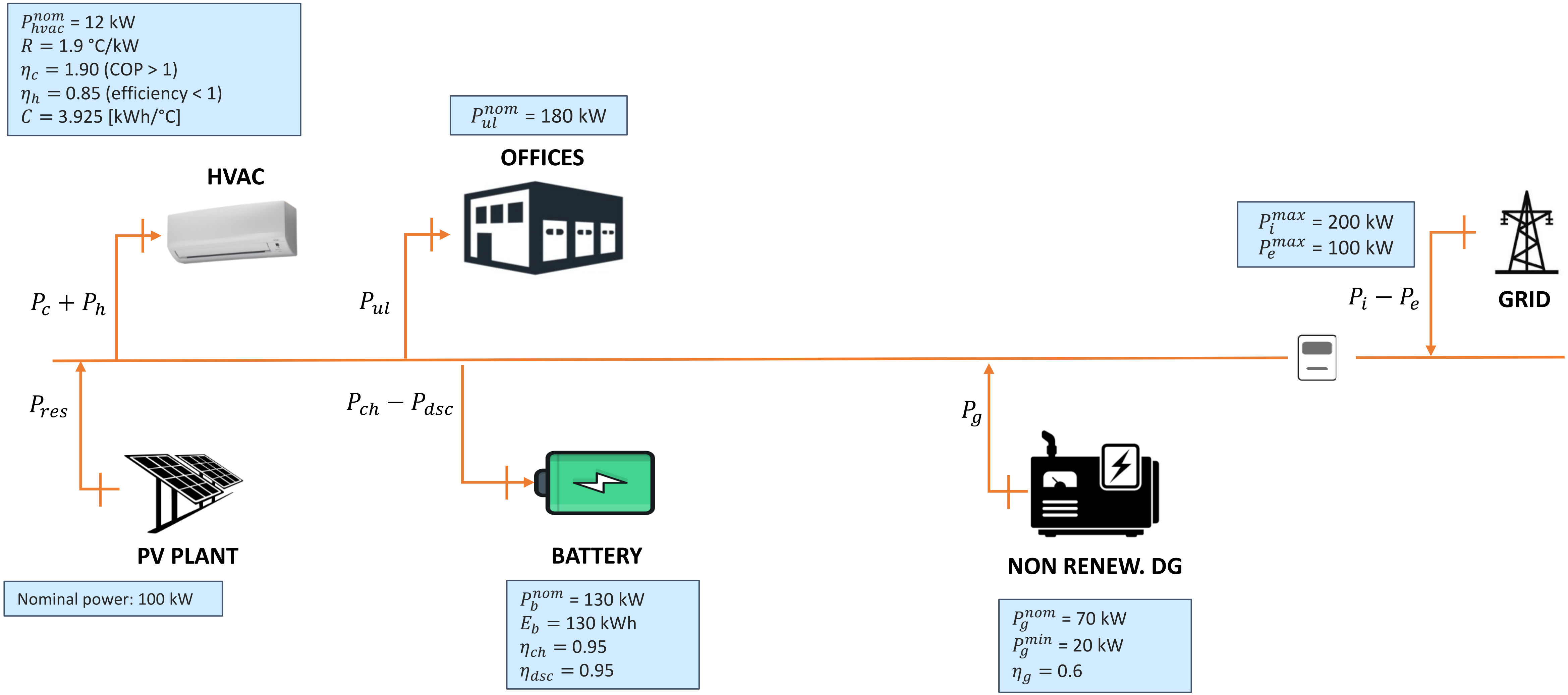
Innovazione e trasformazione digitale
A.A. 2025-26

Modulo 2: Digitalizzazione e ottimizzazione nel settore dell’energia elettrica

Prof. Francesco Conte

Projects

Project 1



P_{hvac}^{nom} = 12 kW
R = 1.9 °C/kW
η_c = 1.90 (COP > 1)
η_h = 0.85 (efficiency < 1)
C = 3.925 [kWh/°C]
HVAC
P_{ul}^{nom} = 180 kW
OFFICES
P_i^{max} = 200 kW
P_e^{max} = 100 kW
GRID
$P_c + P_h$
P_{ul}
$P_i - P_e$
P_{res}
$P_{ch} - P_{dsc}$
P_g
PV PLANT
Nominal power: 100 kW
BATTERY
P_b^{nom} = 130 kW
E_b = 130 kWh
η_{ch} = 0.95
η_{dsc} = 0.95
NON RENEW. DG
P_g^{nom} = 70 kW
P_g^{min} = 20 kW
η_g = 0.6

Project 1

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the temperature around the set-point $T^{sp} = 22°C$ (from April to Sept.) $T^{sp} = 20°C$ (form Oct. to March) and with a tolerance of Δ = 2°C;

c) keep the battery State-of-Charge (SoC) within the interval [0.1   0.9];

Assumptions:

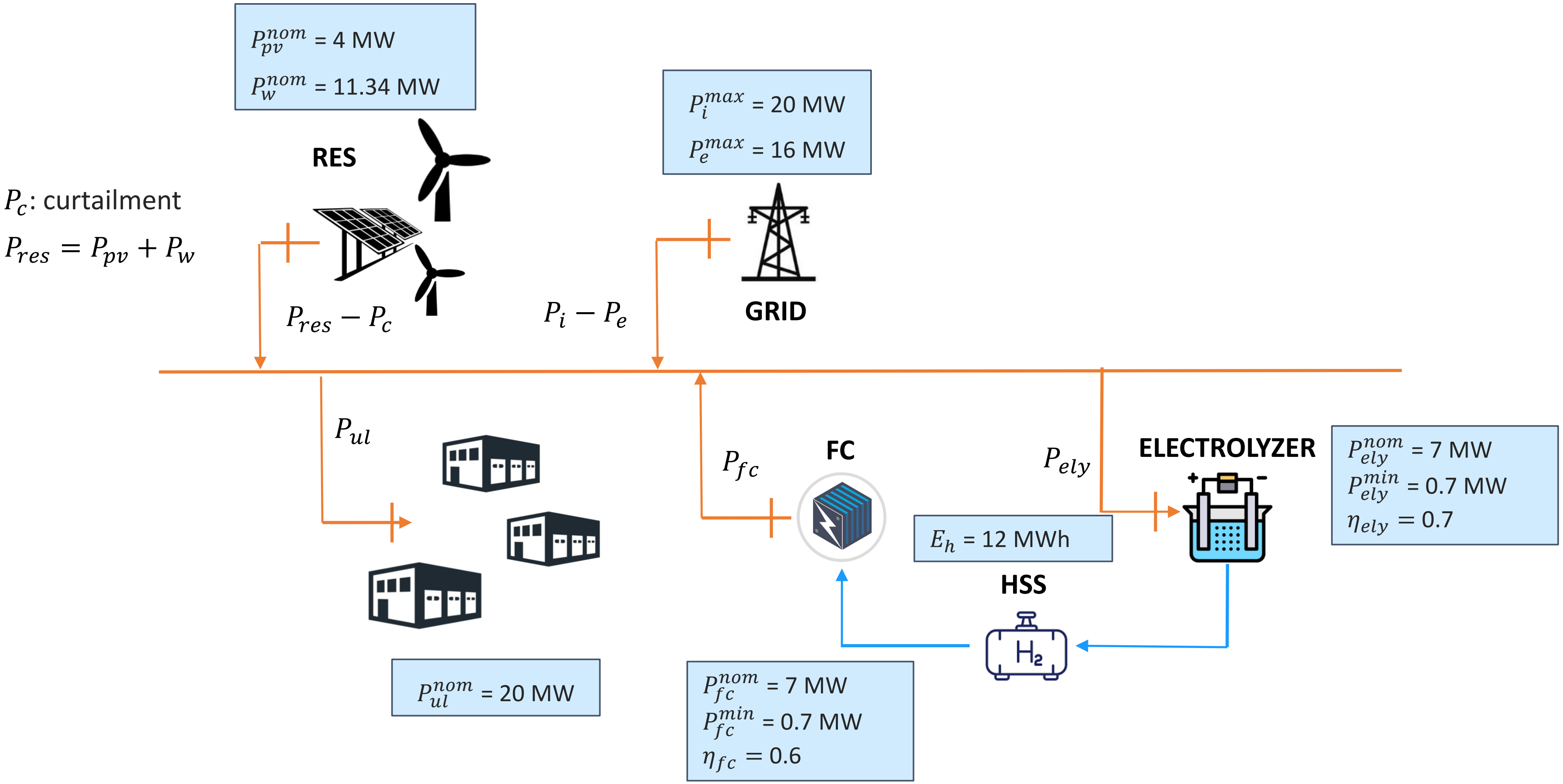
- P_c e P_h can be set in the interval $[0 \quad P_{hvac}^{nom}]$;
- $\Delta t_s = 1$ h;
- $T(k)$, $T_{ex}(k), P_{pv}(k), SoC(k)$, $P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts T_{ex}, P_{pv}, and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 1

Dataset:

- T_{ex}: T_ex_rome_campus_bio_medico_2022.mat
- P_{pv} : Ir_rome_campus_bio_medico_2022.mat
- P_{ul} : office_load.mat
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.45 and with 0.60 €/kWh

Project 2



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Project 2

Control Specifications

Objectives:

a) minimize the cost / maximize the income of the port;

b) satisfy the load of uncontrolled load;

c) minimize the curtailement P_c.

Assumptions:

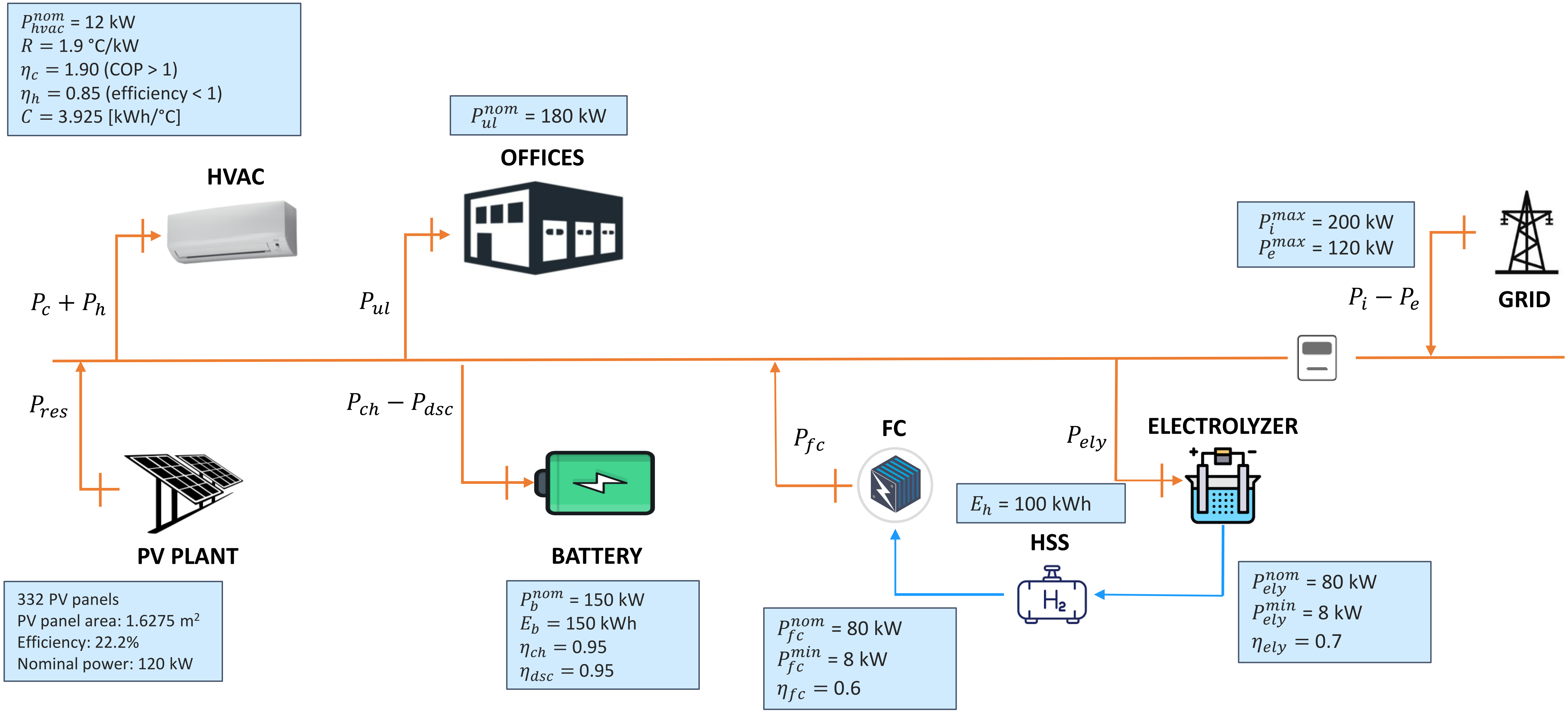
- $\Delta t_s = 1$ h;
- $P_{pv}(k), P_w(k), SoH(k), P_{ul}(k), P_h(k)$ are measured at time k;
- at time k, we have: forecasts of P_{pv}, P_w and P_{ul}
- cost of imported energy $c_I(k)$ is known for the next 24h;
- price of exported energy $p_e(k)$ is known for the next 24h.

Project 2

Dataset:

- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : buildings_load.mat (to be scaled to obtain the right value)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat

Project 3



P_{hvac}^{nom} = 12 kW
R = 1.9 °C/kW
η_c = 1.90 (COP > 1)
η_h = 0.85 (efficiency < 1)
C = 3.925 [kWh/°C]
HVAC
$P_c + P_h$
P_{ul}^{nom} = 180 kW
OFFICES
P_{ul}
P_i^{max} = 200 kW
P_e^{max} = 120 kW
GRID
$P_i - P_e$
P_{res}
PV PLANT
332 PV panels
PV panel area: 1.6275 m^2
Efficiency: 22.2%
Nominal power: 120 kW
$P_{ch} - P_{dsc}$
BATTERY
P_b^{nom} = 150 kW
E_b = 150 kWh
η_{ch} = 0.95
η_{dsc} = 0.95
P_{fc}
FC
P_{fc}^{nom} = 80 kW
P_{fc}^{min} = 8 kW
η_{fc} = 0.6
E_h = 100 kWh
HSS
H_2
P_{ely}
ELECTROLYZER
P_{ely}^{nom} = 80 kW
P_{ely}^{min} = 8 kW
η_{ely} = 0.7

Project 3

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the temperature around the set-point $T^{sp} = 22°C$ (from April to Sept.) $T^{sp} = 20°C$ (form Oct. to March) and with a tolerance of Δ = 2°C;

c) keep the battery State-of-Charge (SoC) within the interval [0.1 0.9].

Assumptions:

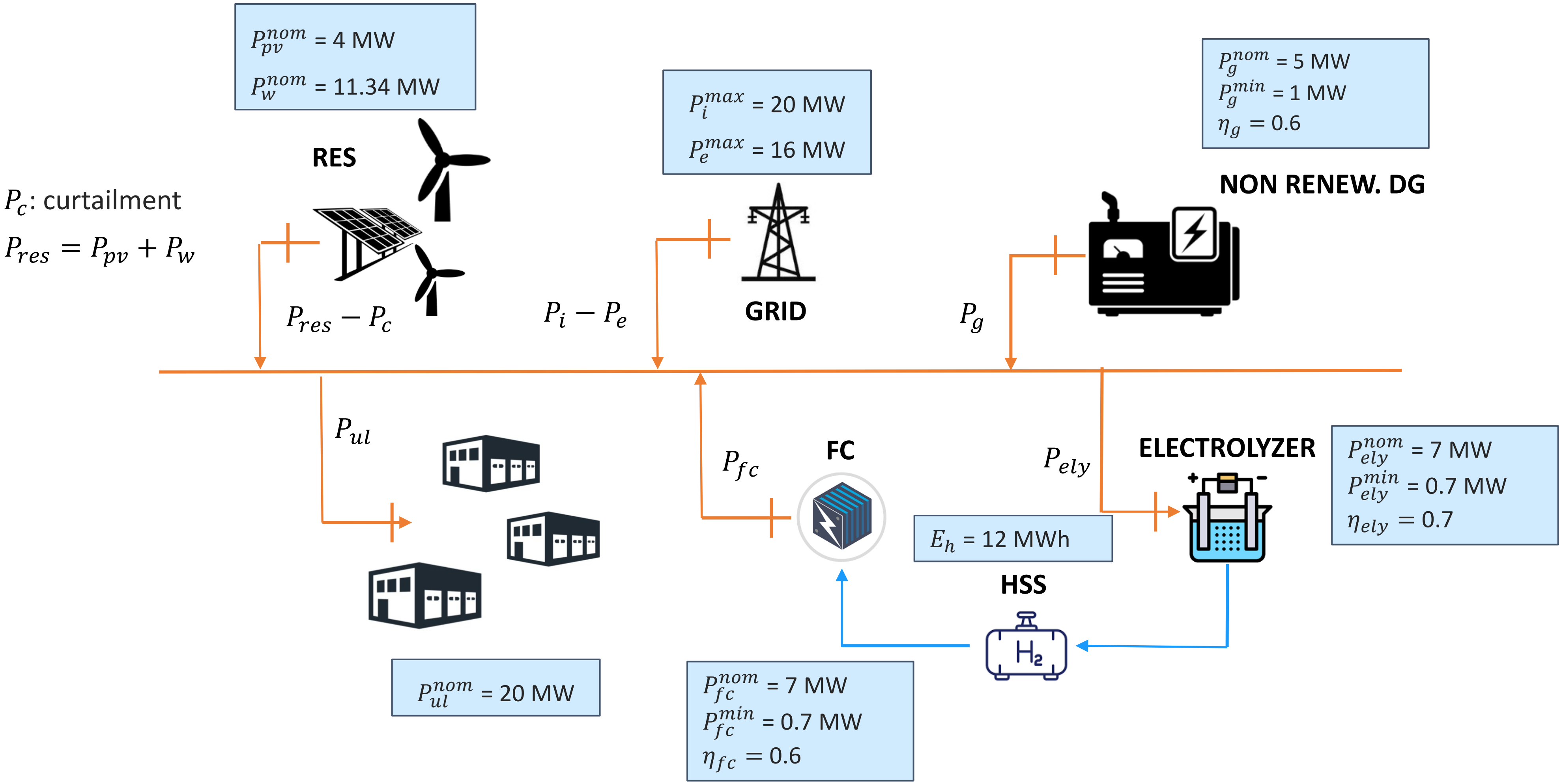
- P_c e P_h can be set in the interval [0 P_{hvac}^{nom}];
- $\Delta t_s = 1$ h;
- $T(k)$, $T_{ex}(k), P_{pv}(k), SoC(k)$, $P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts T_{ex}, P_{pv}, and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN).

Project 3

Dataset:

- T_{ex}: T_ex_rome_campus_bio_medico_2022.mat
- P_{pv} : Ir_rome_campus_bio_medico_2022.mat
- P_{ul} : office_load.mat
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.45 and with 0.60 €/kWh

Project 4



P_{pv}^{nom} = 4 MW
P_{w}^{nom} = 11.34 MW
RES
P_c: curtailment
$P_{res} = P_{pv} + P_w$
$P_{res} - P_c$
P_i^{max} = 20 MW
P_e^{max} = 16 MW
GRID
$P_i - P_e$
P_g^{nom} = 5 MW
P_g^{min} = 1 MW
η_g = 0.6
NON RENEW. DG
P_g
P_{ul}
P_{fc}
FC
P_{ely}
ELECTROLYZER
P_{ely}^{nom} = 7 MW
P_{ely}^{min} = 0.7 MW
η_{ely} = 0.7
E_h = 12 MWh
HSS
H_2
P_{ul}^{nom} = 20 MW
P_{fc}^{nom} = 7 MW
P_{fc}^{min} = 0.7 MW
η_{fc} = 0.6

Project 4

Control specifications

Objectives:

a) minimize the cost / maximize the income of the port;

b) satisfy the load of uncontrolled load;

c) minimize the curtailement P_c.

Assumptions:

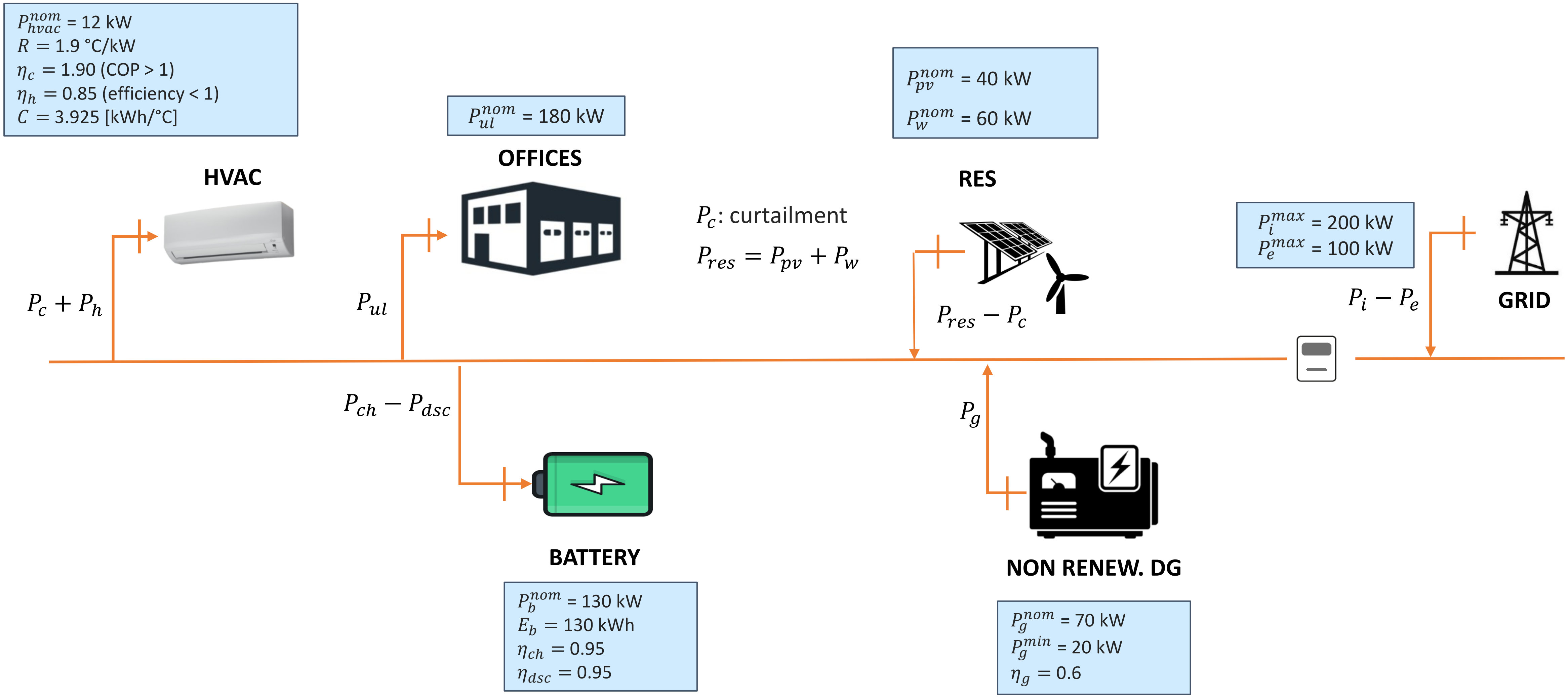
- $\Delta t_s = 1$ h;
- $P_{pv}(k), P_w(k), SoC(k), SoH(k), P_{ul}(k), P_h(k)$ are measured at time k;
- at time k, we have: forecasts of P_{pv}, P_w and P_{ul}
- cost of imported energy $c_l(k)$ is known for the next 24h (we use ARERA prices);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 4

Dataset:

- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : buildings_load.mat (to be scaled to obtain the right value)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.45 and with 0.60 €/kWh

Project 5



P_{hvac}^{nom} = 12 kW
R = 1.9 °C/kW
η_c = 1.90 (COP > 1)
η_h = 0.85 (efficiency < 1)
C = 3.925 [kWh/°C]
HVAC
$P_c + P_h$
P_{ul}^{nom} = 180 kW
OFFICES
P_{ul}
P_c: curtailment
$P_{res} = P_{pv} + P_w$
P_{pv}^{nom} = 40 kW
P_w^{nom} = 60 kW
RES
$P_{res} - P_c$
P_i^{max} = 200 kW
P_e^{max} = 100 kW
$P_i - P_e$
GRID
$P_{ch} - P_{dsc}$
BATTERY
P_b^{nom} = 130 kW
E_b = 130 kWh
η_{ch} = 0.95
η_{dsc} = 0.95
P_g
NON RENEW. DG
P_g^{nom} = 70 kW
P_g^{min} = 20 kW
η_g = 0.6

Project 5

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the temperature around the set-point $T^{sp} = 22°C$ (from April to Sept.) $T^{sp} = 20°C$ (form Oct. to March) and with a tolerance of Δ = 2°C;

c) keep the battery State-of-Charge (SoC) within the interval [0.1 0.9];

Assumptions:

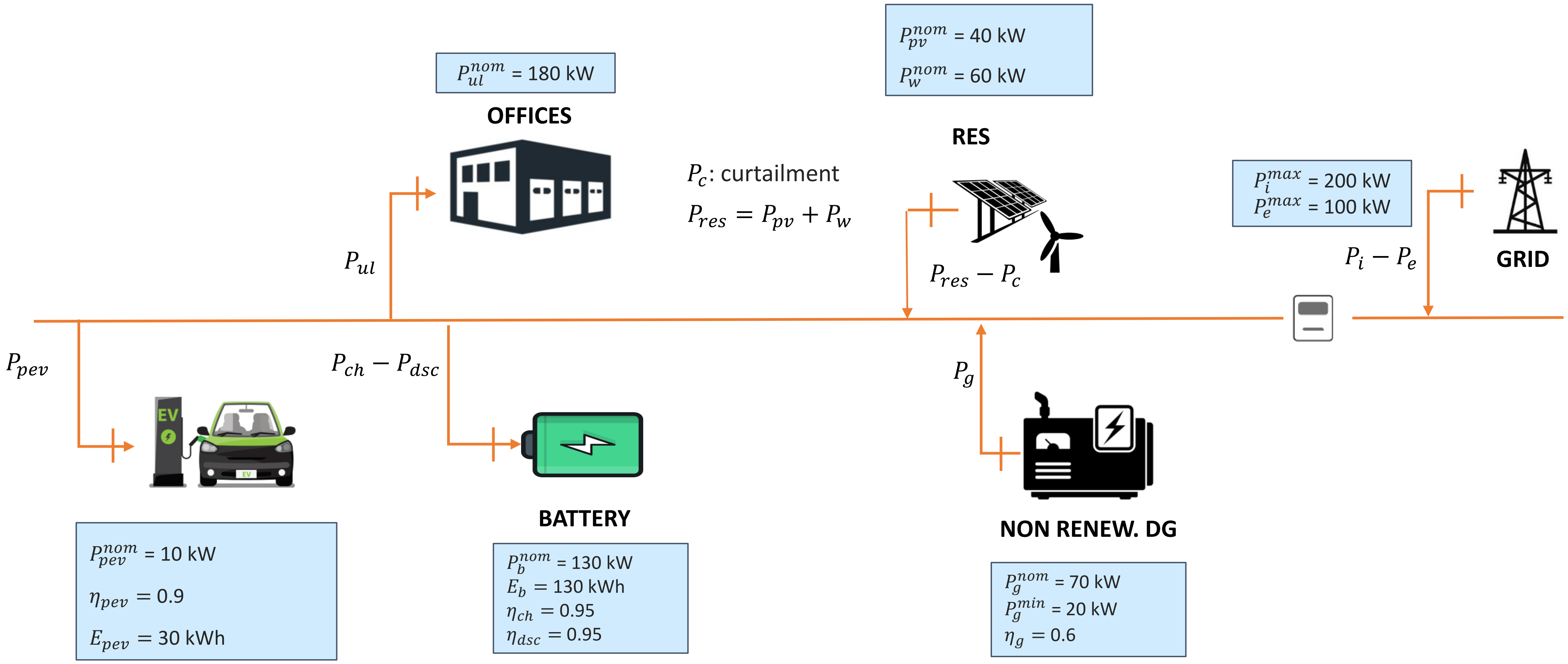
- P_c e P_h can be set in the interval $[0 \quad P_{hvac}^{nom}]$;
- $\Delta t_s = 1$ h;
- $T(k)$, $T_{ex}(k), P_{pv}(k), P_w(k), SoC(k)$, $P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts T_{ex}, P_{pv}, P_w and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 5

Dataset:

- T_{ex}: T_ex_rome_campus_bio_medico_2022.mat
- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : office_load.mat
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.45 and with 0.60 €/kWh

Project 6



P_{ul}^{nom} = 180 kW
OFFICES
P_{ul}
P_c: curtailment
$P_{res} = P_{pv} + P_w$
P_{pv}^{nom} = 40 kW
P_w^{nom} = 60 kW
RES
$P_{res} - P_c$
P_i^{max} = 200 kW
P_e^{max} = 100 kW
$P_i - P_e$
GRID
P_{pev}
EV
P_{pev}^{nom} = 10 kW
$\eta_{pev} = 0.9$
E_{pev} = 30 kWh
$P_{ch} - P_{dsc}$
BATTERY
P_b^{nom} = 130 kW
E_b = 130 kWh
$\eta_{ch} = 0.95$
$\eta_{dsc} = 0.95$
P_g
NON RENEW. DG
P_g^{nom} = 70 kW
P_g^{min} = 20 kW
$\eta_g = 0.6$

Project 6

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the battery State-of-Charge (SoC) within the interval [0.1   0.9];

c) recharge the PEV when required

Assumptions:

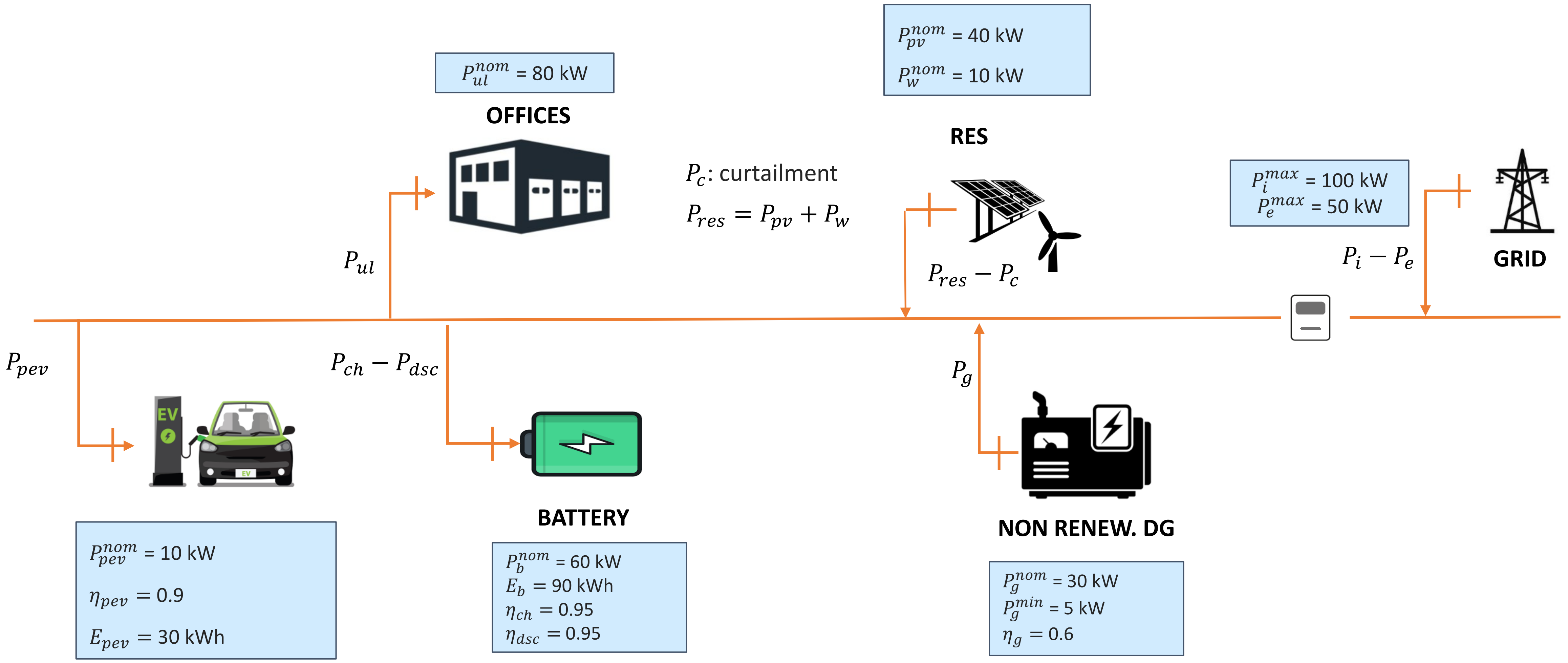
- P_c e P_h can be set in the interval $[0 \quad P_{hvac}^{nom}]$;
- $\Delta t_s = 1$ h;
- $T(k), P_{pv}(k), P_w(k), SoC(k), P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts P_{pv}, P_w and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 6

Dataset:

- T_{ex}: T_ex_rome_campus_bio_medico_2022.mat
- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : office_load.mat
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.45 and with 0.60 €/kWh

Project 7



P_{ul}^{nom} = 80 kW
OFFICES
P_c: curtailment
$P_{res} = P_{pv} + P_w$
P_{pv}^{nom} = 40 kW
P_w^{nom} = 10 kW
RES
P_i^{max} = 100 kW
P_e^{max} = 50 kW
GRID
P_{ul}
$P_{res} - P_c$
$P_i - P_e$
P_{pev}
$P_{ch} - P_{dsc}$
P_g
EV
BATTERY
NON RENEW. DG
P_{pev}^{nom} = 10 kW
$\eta_{pev} = 0.9$
$E_{pev} = 30$ kWh
P_b^{nom} = 60 kW
$E_b = 90$ kWh
$\eta_{ch} = 0.95$
$\eta_{dsc} = 0.95$
P_g^{nom} = 30 kW
P_g^{min} = 5 kW
$\eta_g = 0.6$

Project 7

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the battery State-of-Charge (SoC) within the interval [0.1 0.9];

c) recharge the PEV when required

Assumptions:

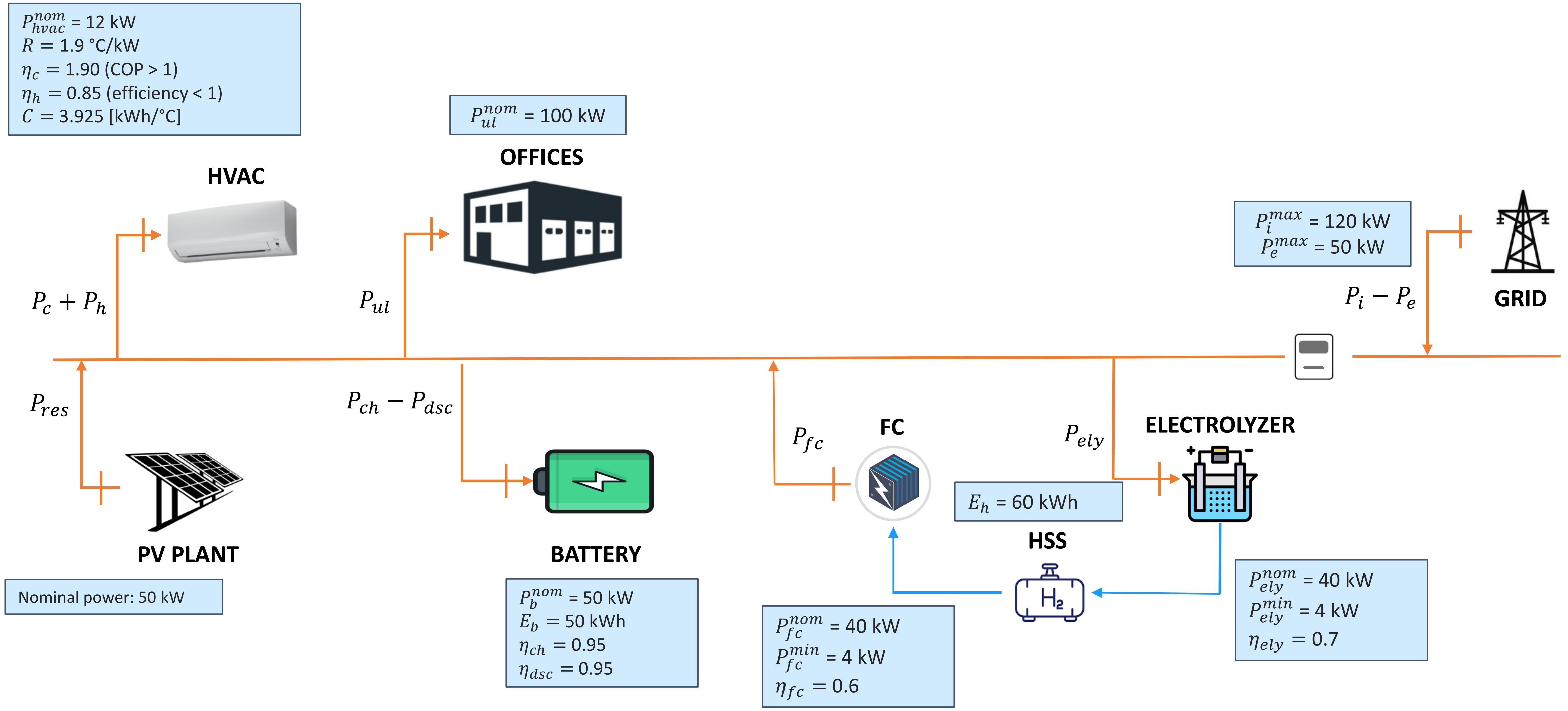
- P_c e P_h can be set in the interval [0 P_{hvac}^{nom}];
- $\Delta t_s = 1$ h;
- $T(k), P_{pv}(k), P_w(k), SoC(k), P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts P_{pv}, P_w and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 7

Dataset:

- T_{ex}: T_ex_rome_campus_bio_medico_2022.mat
- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : office_load.mat (scale the value for obtaining the correct nominal power)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.45 and with 0.60 €/kWh

Project 8



P_{hvac}^{nom} = 12 kW
R = 1.9 °C/kW
η_c = 1.90 (COP > 1)
η_h = 0.85 (efficiency < 1)
C = 3.925 [kWh/°C]
HVAC
$P_c + P_h$
P_{ul}^{nom} = 100 kW
OFFICES
P_{ul}
P_i^{max} = 120 kW
P_e^{max} = 50 kW
$P_i - P_e$
GRID
P_{res}
PV PLANT
Nominal power: 50 kW
$P_{ch} - P_{dsc}$
BATTERY
P_b^{nom} = 50 kW
E_b = 50 kWh
η_{ch} = 0.95
η_{dsc} = 0.95
P_{fc}
FC
P_{fc}^{nom} = 40 kW
P_{fc}^{min} = 4 kW
η_{fc} = 0.6
P_{ely}
ELECTROLYZER
E_h = 60 kWh
HSS
H_2
P_{ely}^{nom} = 40 kW
P_{ely}^{min} = 4 kW
η_{ely} = 0.7

Project 8

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the temperature around the set-point $T^{sp} = 22°C$ (from April to Sept.) $T^{sp} = 20°C$ (form Oct. to March) and with a tolerance of Δ = 2°C;

c) keep the battery State-of-Charge (SoC) within the interval [0.1 0.9].

Assumptions:

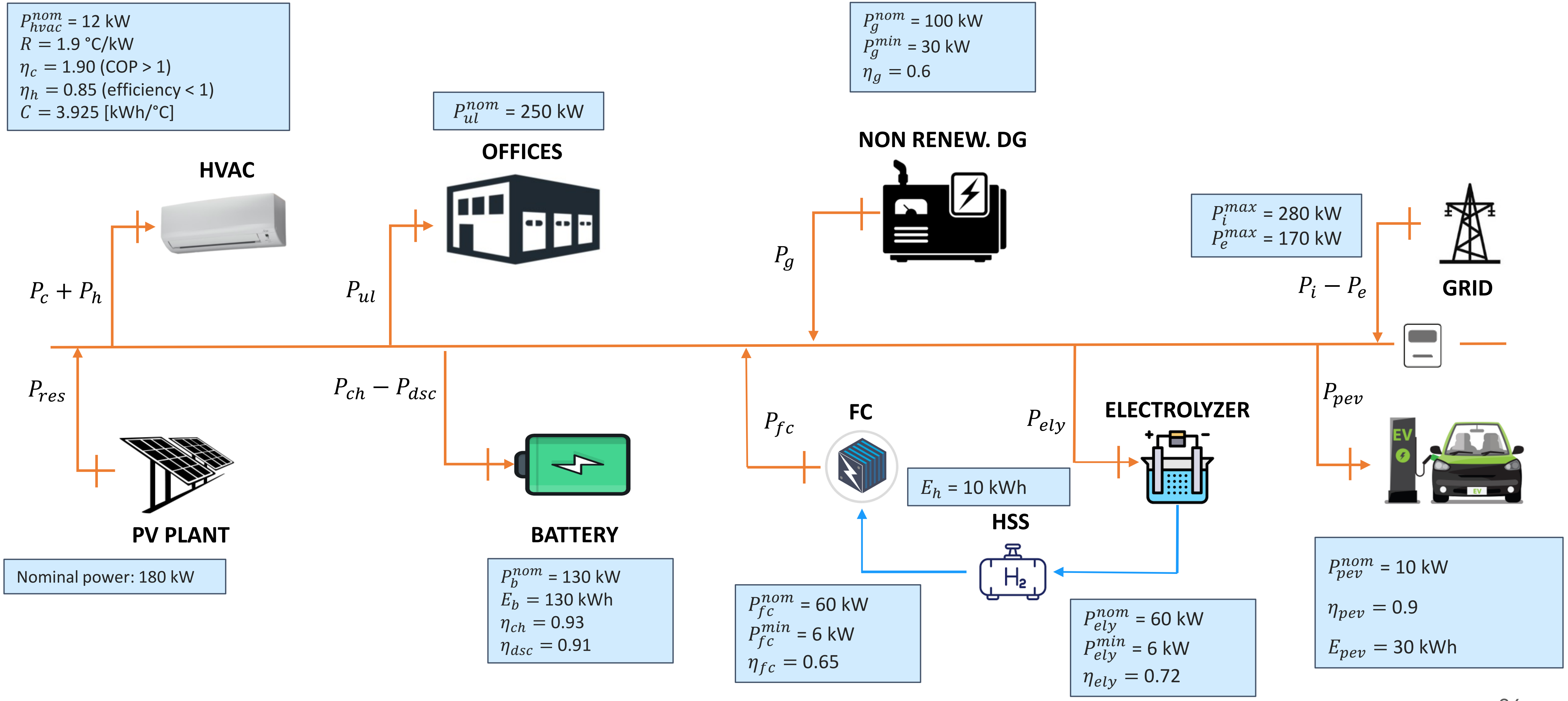
- P_c e P_h can be set in the interval [0 P_{hvac}^{nom}];
- $\Delta t_s = 1$ h;
- $T(k)$, $T_{ex}(k), P_{pv}(k), SoC(k)$, $P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts T_{ex}, P_{pv}, and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN).

Project 8

Dataset:

- T_{ex}: T_ex_rome_campus_bio_medico_2022.mat
- P_{pv} : Ir_rome_campus_bio_medico_2022.mat
- P_{ul} : office_load.mat (to be scaled to obtain the right nominal power)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.45 and with 0.60 €/kWh

Project 9



P_{hvac}^{nom} = 12 kW
R = 1.9 °C/kW
η_c = 1.90 (COP > 1)
η_h = 0.85 (efficiency < 1)
C = 3.925 [kWh/°C]
HVAC
$P_c + P_h$
P_{ul}^{nom} = 250 kW
OFFICES
P_{ul}
P_g^{nom} = 100 kW
P_g^{min} = 30 kW
η_g = 0.6
NON RENEW. DG
P_g
P_i^{max} = 280 kW
P_e^{max} = 170 kW
$P_i - P_e$
GRID
P_{res}
PV PLANT
Nominal power: 180 kW
$P_{ch} - P_{dsc}$
BATTERY
P_b^{nom} = 130 kW
E_b = 130 kWh
η_{ch} = 0.93
η_{dsc} = 0.91
P_{fc}
FC
P_{fc}^{nom} = 60 kW
P_{fc}^{min} = 6 kW
η_{fc} = 0.65
E_h = 10 kWh
HSS
H_2
P_{ely}
ELECTROLYZER
P_{ely}^{nom} = 60 kW
P_{ely}^{min} = 6 kW
η_{ely} = 0.72
P_{pev}
EV
P_{pev}^{nom} = 10 kW
η_{pev} = 0.9
E_{pev} = 30 kWh

Project 9

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the temperature around the set-point $T^{sp} = 22°C$ (from April to Sept.) $T^{sp} = 20°C$ (form Oct. to March) and with a tolerance of Δ = 2°C;

c) keep the battery State-of-Charge (SoC) within the interval [0.1 0.9];

d) recharge the PEV when required

Assumptions:

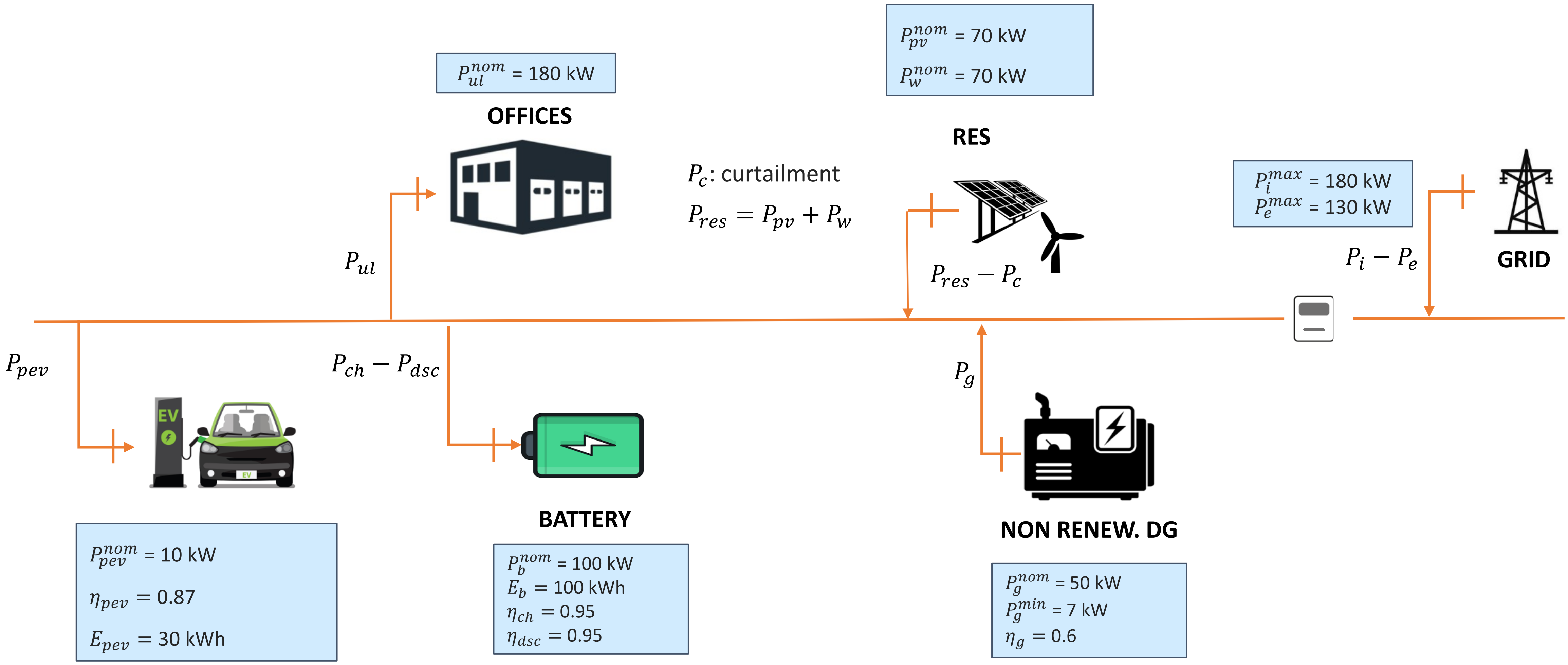
- P_c e P_h can be set in the interval [0 P_{hvac}^{nom}];
- $\Delta t_s = 1$ h;
- $T(k)$, $T_{ex}(k), P_{pv}(k), SoC(k)$, $P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts T_{ex}, P_{pv}, and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 9

Dataset:

- T_{ex}: T_ex_rome_campus_bio_medico_2022.mat
- P_{pv} : Ir_rome_campus_bio_medico_2022.mat
- P_{ul} : office_load.mat (scale the value to obtain the right nominal value)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.15 and with 0.45 €/kWh

Project 10



P_{ul}^{nom} = 180 kW
OFFICES
P_c: curtailment
$P_{res} = P_{pv} + P_w$
P_{pv}^{nom} = 70 kW
P_w^{nom} = 70 kW
RES
P_i^{max} = 180 kW
P_e^{max} = 130 kW
GRID
P_{ul}
$P_{res} - P_c$
$P_i - P_e$
P_{pev}
$P_{ch} - P_{dsc}$
P_g
EV
BATTERY
NON RENEW. DG
P_{pev}^{nom} = 10 kW
$\eta_{pev} = 0.87$
E_{pev} = 30 kWh
P_b^{nom} = 100 kW
E_b = 100 kWh
$\eta_{ch} = 0.95$
$\eta_{dsc} = 0.95$
P_g^{nom} = 50 kW
P_g^{min} = 7 kW
$\eta_g = 0.6$

Project 10

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the battery State-of-Charge (SoC) within the interval [0.1   0.9];

c) recharge the PEV when required

Assumptions:

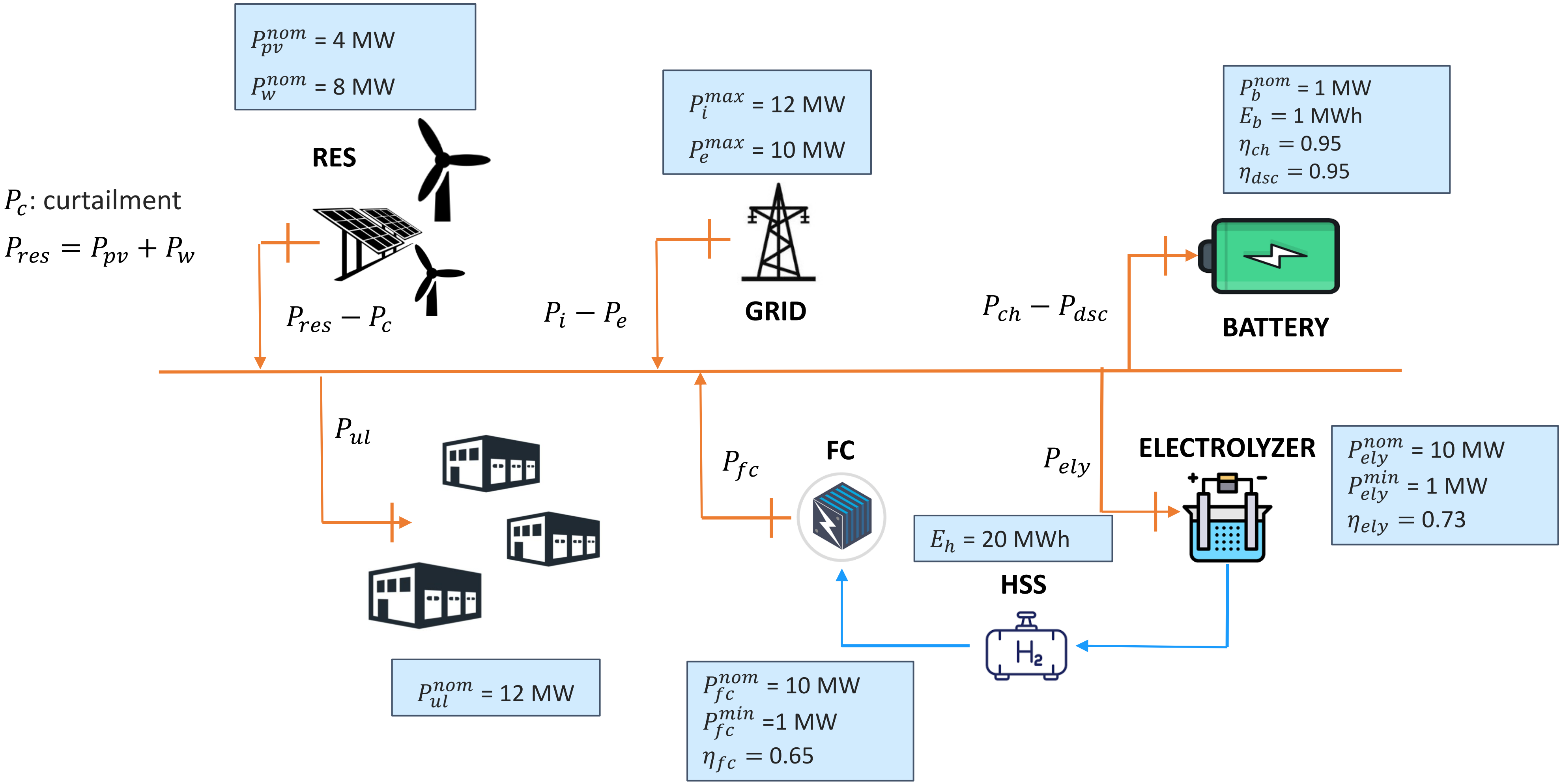
- $\Delta t_s = 1$ h;
- $T(k), P_{pv}(k), P_w(k), SoC(k), P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts P_{pv}, P_w and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 10

Dataset:

- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : office_load.mat (scale the value for obtaining the correct nominal power)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.15 and with 0.45 €/kWh

Project 11



P_{pv}^{nom} = 4 MW
P_{w}^{nom} = 8 MW
RES
P_c: curtailment
$P_{res} = P_{pv} + P_w$
$P_{res} - P_c$
P_i^{max} = 12 MW
P_e^{max} = 10 MW
GRID
$P_i - P_e$
P_b^{nom} = 1 MW
E_b = 1 MWh
η_{ch} = 0.95
η_{dsc} = 0.95
BATTERY
$P_{ch} - P_{dsc}$
P_{ul}
P_{ul}^{nom} = 12 MW
P_{fc}
FC
P_{fc}^{nom} = 10 MW
P_{fc}^{min} =1 MW
η_{fc} = 0.65
E_h = 20 MWh
HSS
H_2
P_{ely}
ELECTROLYZER
P_{ely}^{nom} = 10 MW
P_{ely}^{min} = 1 MW
η_{ely} = 0.73

Project 11

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial system;

b) satisfy the load of uncontrolled load;

c) keep the battery SoC between 10% to 90%

d) minimize the curtailement P_c.

Assumptions:

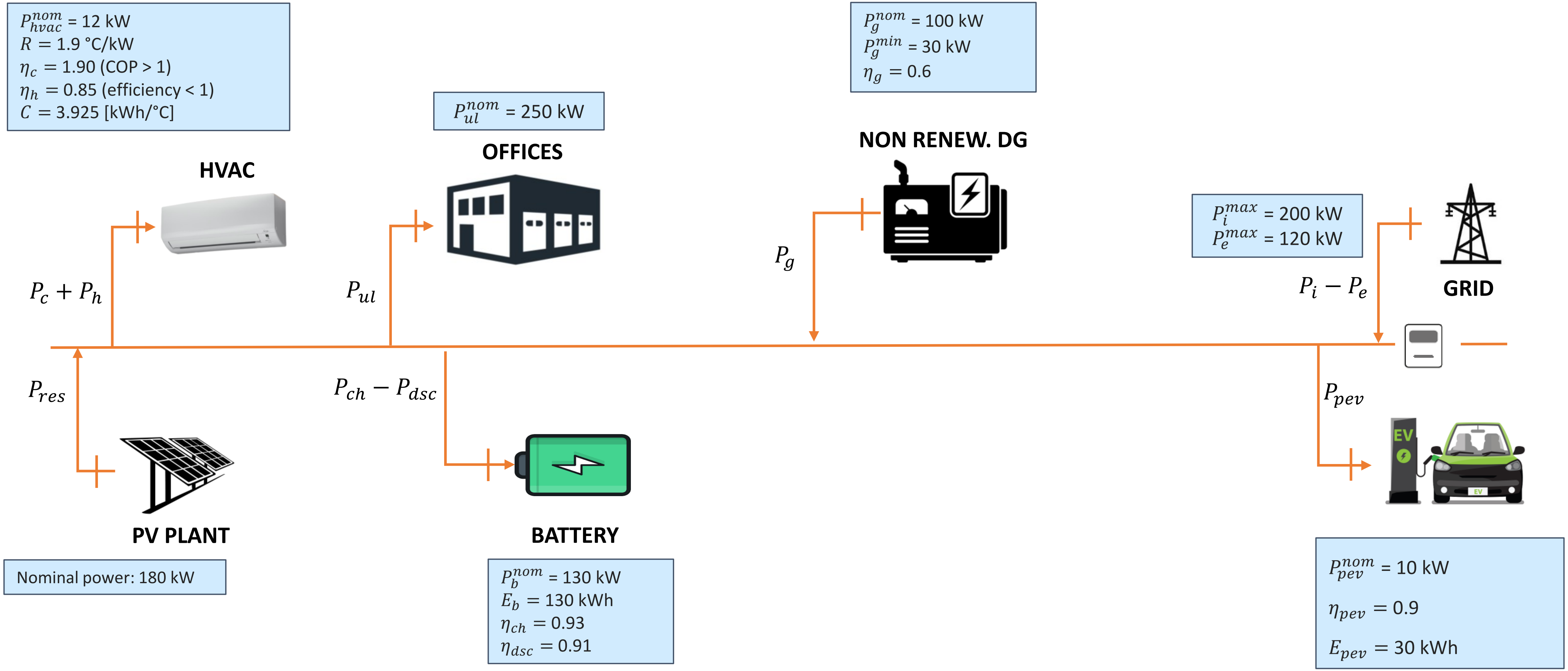
- $\Delta t_s = 1$ h;
- $P_{pv}(k), P_w(k), SoC(k), SoH(k), P_{ul}(k), P_h(k)$ are measured at time k;
- at time k, we have: forecasts of P_{pv}, P_w and P_{ul}
- cost of imported energy $c_l(k)$ is known for the next 24h;
- price of exported energy $p_e(k)$ is known for the next 24h.

Project 11

Dataset:

- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : buildings_load.mat (to be scaled to obtain the right value)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat

Project 12


P_{hvac}^{nom} = 12 kW
R = 1.9 °C/kW
η_c = 1.90 (COP > 1)
η_h = 0.85 (efficiency < 1)
C = 3.925 [kWh/°C]
HVAC
$P_c + P_h$
P_{ul}^{nom} = 250 kW
OFFICES
P_{ul}
P_g^{nom} = 100 kW
P_g^{min} = 30 kW
η_g = 0.6
NON RENEW. DG
P_g
P_i^{max} = 200 kW
P_e^{max} = 120 kW
$P_i - P_e$
GRID
P_{res}
PV PLANT
Nominal power: 180 kW
$P_{ch} - P_{dsc}$
BATTERY
P_b^{nom} = 130 kW
E_b = 130 kWh
η_{ch} = 0.93
η_{dsc} = 0.91
P_{pev}
EV
P_{pev}^{nom} = 10 kW
η_{pev} = 0.9
E_{pev} = 30 kWh

Project 12

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the battery State-of-Charge (SoC) within the interval $[0.05 \quad 0.95]$;

c) recharge the PEV when required

Assumptions:

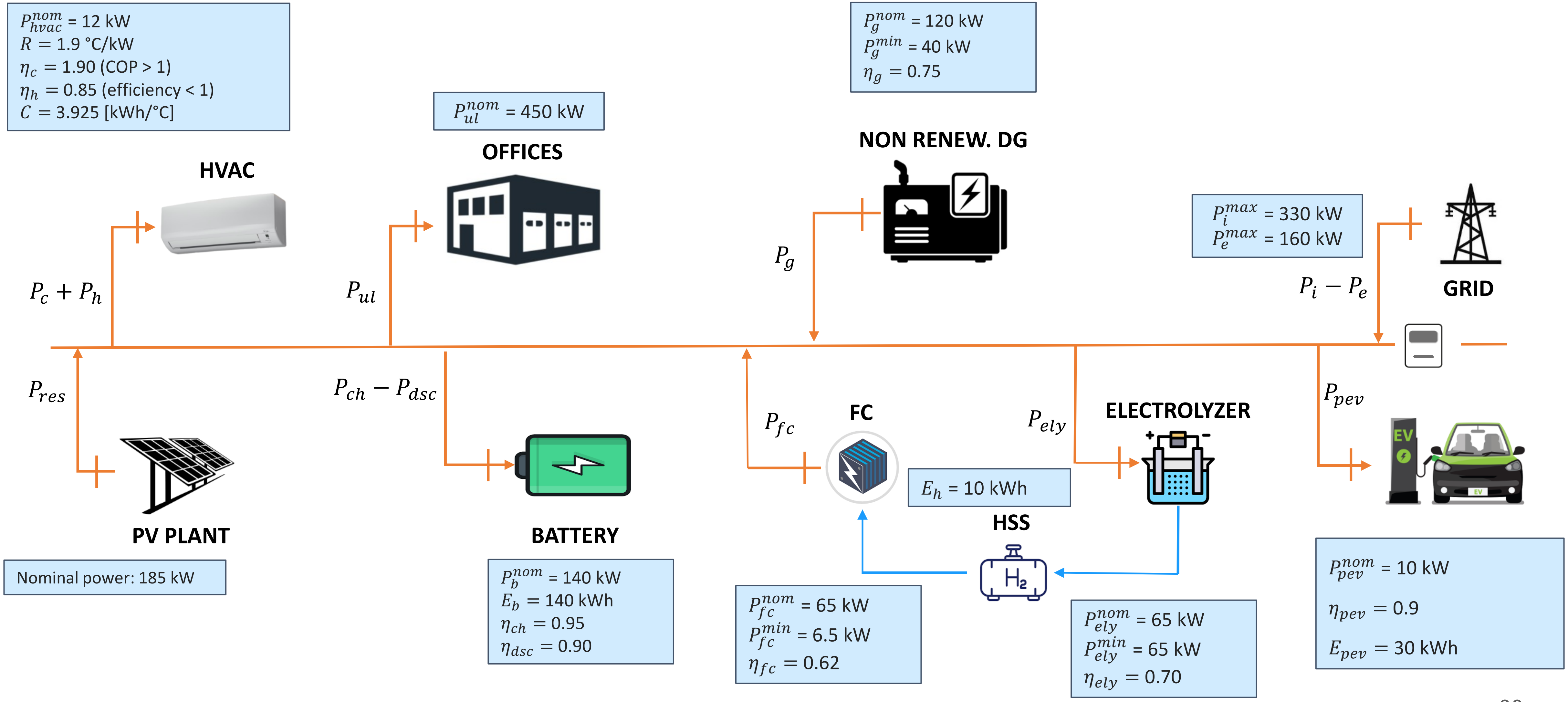
- $\Delta t_s = 1$ h;
- $T(k)$, $T_{ex}(k), P_{pv}(k), SoC(k)$, $P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts T_{ex}, P_{pv}, and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 12

Dataset:

- P_{pv} : Ir_rome_campus_bio_medico_2022.mat
- P_{ul} : office_load.mat (scale the value to obtain the right nominal value)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.15 and with 0.45 €/kWh

Project 13


P_{hvac}^{nom} = 12 kW
R = 1.9 °C/kW
η_c = 1.90 (COP > 1)
η_h = 0.85 (efficiency < 1)
C = 3.925 [kWh/°C]
HVAC
P_{ul}^{nom} = 450 kW
OFFICES
P_g^{nom} = 120 kW
P_g^{min} = 40 kW
η_g = 0.75
NON RENEW. DG
P_i^{max} = 330 kW
P_e^{max} = 160 kW
GRID
$P_c + P_h$
P_{ul}
P_g
$P_i - P_e$
P_{res}
$P_{ch} - P_{dsc}$
P_{fc}
FC
P_{ely}
ELECTROLYZER
P_{pev}
EV
E_h = 10 kWh
HSS
H_2
PV PLANT
Nominal power: 185 kW
BATTERY
P_b^{nom} = 140 kW
E_b = 140 kWh
η_{ch} = 0.95
η_{dsc} = 0.90
P_{fc}^{nom} = 65 kW
P_{fc}^{min} = 6.5 kW
η_{fc} = 0.62
P_{ely}^{nom} = 65 kW
P_{ely}^{min} = 65 kW
η_{ely} = 0.70
P_{pev}^{nom} = 10 kW
η_{pev} = 0.9
E_{pev} = 30 kWh

Project 13

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the temperature around the set-point $T^{sp} = 22°C$ (from April to Sept.) $T^{sp} = 20°C$ (form Oct. to March) and with a tolerance of Δ = 2°C;

c) keep the battery State-of-Charge (SoC) within the interval [0.1 0.9];

d) recharge the PEV when required

Assumptions:

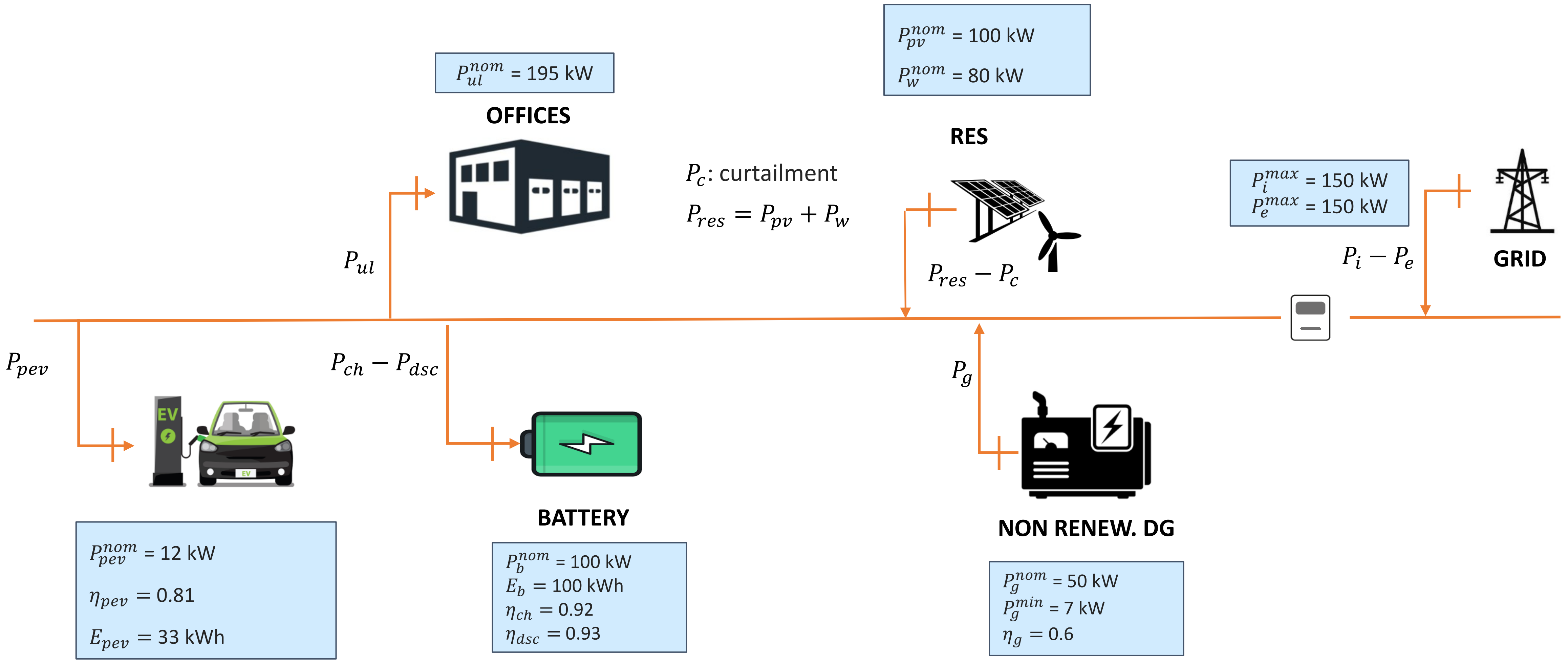
- P_c e P_h can be set in the interval [0 P_{hvac}^{nom}];
- $\Delta t_s = 1$ h;
- $T(k)$, $T_{ex}(k), P_{pv}(k), SoC(k)$, $P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts T_{ex}, P_{pv}, and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 13

Dataset:

- T_{ex}: T_ex_rome_campus_bio_medico_2022.mat
- P_{pv} : Ir_rome_campus_bio_medico_2022.mat
- P_{ul} : office_load.mat (scale the value to obtain the right nominal value)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.15 and with 0.45 €/kWh

Project 14



P_{ul}^{nom} = 195 kW
OFFICES
P_{pv}^{nom} = 100 kW
P_{w}^{nom} = 80 kW
RES
P_c: curtailment
$P_{res} = P_{pv} + P_w$
P_i^{max} = 150 kW
P_e^{max} = 150 kW
GRID
P_{ul}
$P_{res} - P_c$
$P_i - P_e$
P_{pev}
$P_{ch} - P_{dsc}$
P_g
EV
BATTERY
NON RENEW. DG
P_{pev}^{nom} = 12 kW
$\eta_{pev} = 0.81$
$E_{pev} = 33$ kWh
P_b^{nom} = 100 kW
$E_b = 100$ kWh
$\eta_{ch} = 0.92$
$\eta_{dsc} = 0.93$
P_g^{nom} = 50 kW
P_g^{min} = 7 kW
$\eta_g = 0.6$

Project 14

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the battery State-of-Charge (SoC) within the interval [0.1   0.9];

c) recharge the PEV when required

Assumptions:

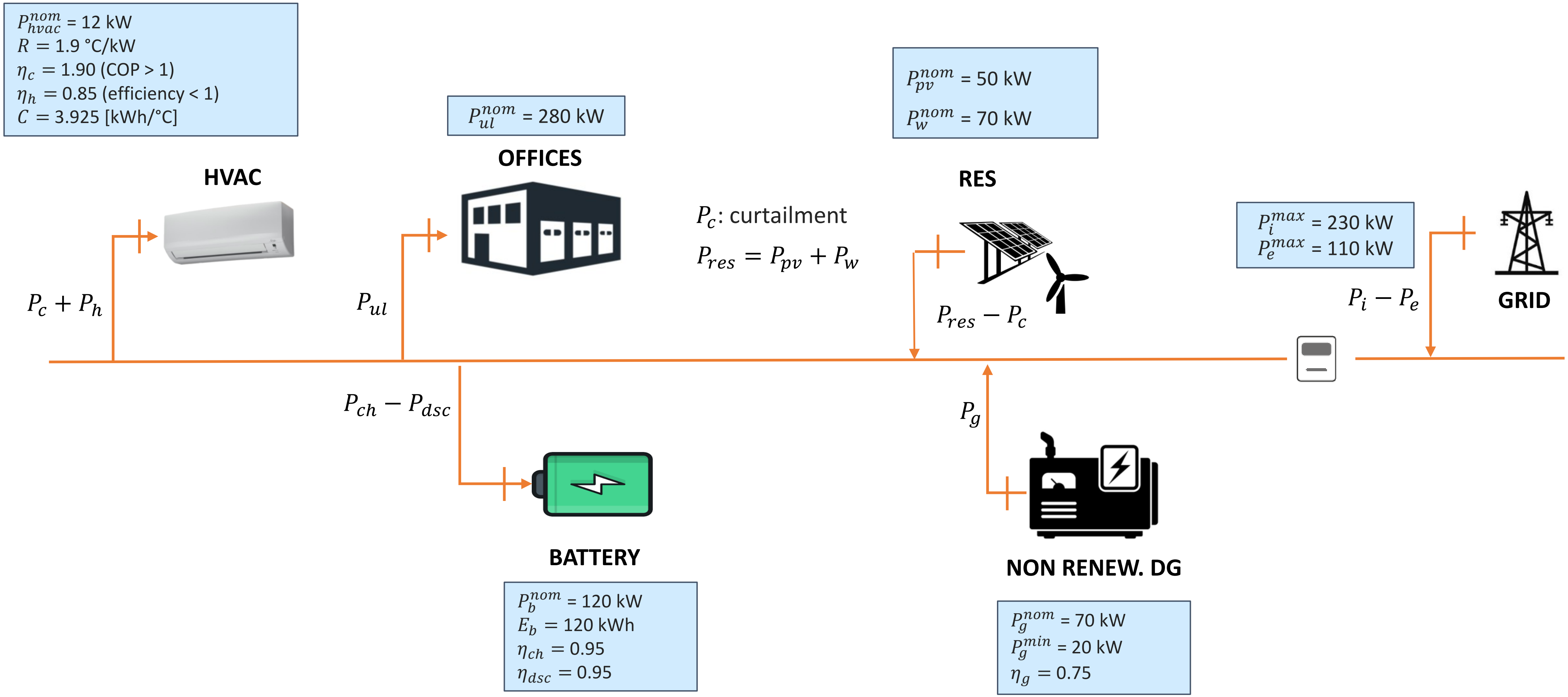
- $\Delta t_s = 1$ h;
- $T(k), P_{pv}(k), P_w(k), SoC(k), P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts P_{pv}, P_w and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 14

Dataset:

- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : office_load.mat (scale the value for obtaining the correct nominal power)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.15 and with 0.45 €/kWh

Project 15


P_{hvac}^{nom} = 12 kW
R = 1.9 °C/kW
η_c = 1.90 (COP > 1)
η_h = 0.85 (efficiency < 1)
C = 3.925 [kWh/°C]
HVAC
$P_c + P_h$
P_{ul}^{nom} = 280 kW
OFFICES
P_{ul}
P_c: curtailment
$P_{res} = P_{pv} + P_w$
P_{pv}^{nom} = 50 kW
P_w^{nom} = 70 kW
RES
$P_{res} - P_c$
P_i^{max} = 230 kW
P_e^{max} = 110 kW
$P_i - P_e$
GRID
$P_{ch} - P_{dsc}$
BATTERY
P_b^{nom} = 120 kW
E_b = 120 kWh
η_{ch} = 0.95
η_{dsc} = 0.95
P_g
NON RENEW. DG
P_g^{nom} = 70 kW
P_g^{min} = 20 kW
η_g = 0.75

Project 15

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial district;

b) keep the temperature around the set-point $T^{sp} = 22°C$ (from April to Sept.) $T^{sp} = 20°C$ (form Oct. to March) and with a tolerance of Δ = 2°C;

c) keep the battery State-of-Charge (SoC) within the interval [0.1 0.9];

Assumptions:

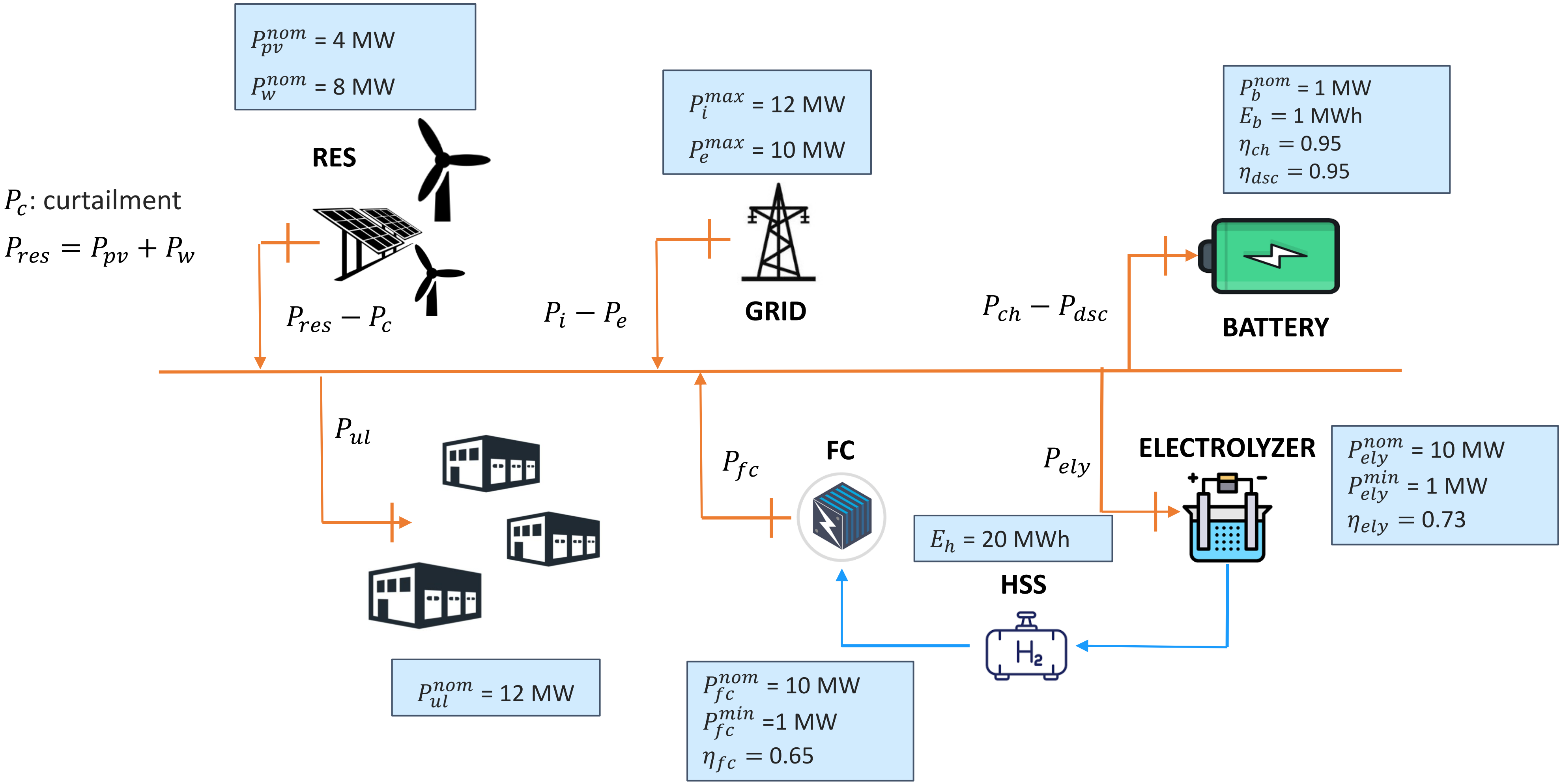
- P_c e P_h can be set in the interval [0 P_{hvac}^{nom}];
- $\Delta t_s = 1$ h;
- $T(k)$, $T_{ex}(k), P_{pv}(k), P_w(k), SoC(k)$, $P_{ul}(k)$, are measured at time k;
- at time k we are provided with the forecasts T_{ex}, P_{pv}, P_w and P_{ul} for the next 24h;
- cost of imported energy $c_l(k)$ is known for the next 24h (see next slide);
- price of exported energy $p_e(k)$ is known for the next 24h (we use PUN);
- price of fuel for the non renewable DG c_f is constant and known.

Project 15

Dataset:

- T_{ex}: T_ex_rome_campus_bio_medico_2022.mat
- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : office_load.mat
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat
- c_f: test with 0.25 and with 0.45 €/kWh

Project 16


P_{pv}^{nom} = 4 MW
P_{w}^{nom} = 8 MW
RES
P_c: curtailment
$P_{res} = P_{pv} + P_w$
$P_{res} - P_c$
P_i^{max} = 12 MW
P_e^{max} = 10 MW
GRID
$P_i - P_e$
P_b^{nom} = 1 MW
E_b = 1 MWh
η_{ch} = 0.95
η_{dsc} = 0.95
BATTERY
$P_{ch} - P_{dsc}$
P_{ul}
P_{ul}^{nom} = 12 MW
P_{fc}
FC
P_{fc}^{nom} = 10 MW
P_{fc}^{min} =1 MW
η_{fc} = 0.65
E_h = 20 MWh
HSS
P_{ely}
ELECTROLYZER
P_{ely}^{nom} = 10 MW
P_{ely}^{min} = 1 MW
η_{ely} = 0.73

Project 16

Control specifications

Objectives:

a) minimize the cost / maximize the income of the industrial system;

b) satisfy the load of uncontrolled load;

c) keep the battery SoC between 10% to 90%

d) minimize the curtailement P_c.

Assumptions:

- $\Delta t_s = 1$ h;
- $P_{pv}(k), P_w(k), SoC(k), SoH(k), P_{ul}(k), P_h(k)$ are measured at time k;
- at time k, we have: forecasts of P_{pv}, P_w and P_{ul}
- cost of imported energy $c_l(k)$ is known for the next 24h;
- price of exported energy $p_e(k)$ is known for the next 24h.

Project 16

Dataset:

- P_{pv}, P_w : res_1_year_pu.mat (pu values to be multiplied into the indicated nominal values)
- P_{ul} : buildings_load.mat (to be scaled to obtain the right value)
- c_l : F1 = 0,53276 F2=0,54858 F3=0,46868 [€/kWh]
- p_e : PUN_2022.mat

Innovazione e trasformazione digitale

f.conte@unicampus.it